Help Menu

The commands are described under the **Help Menu**.

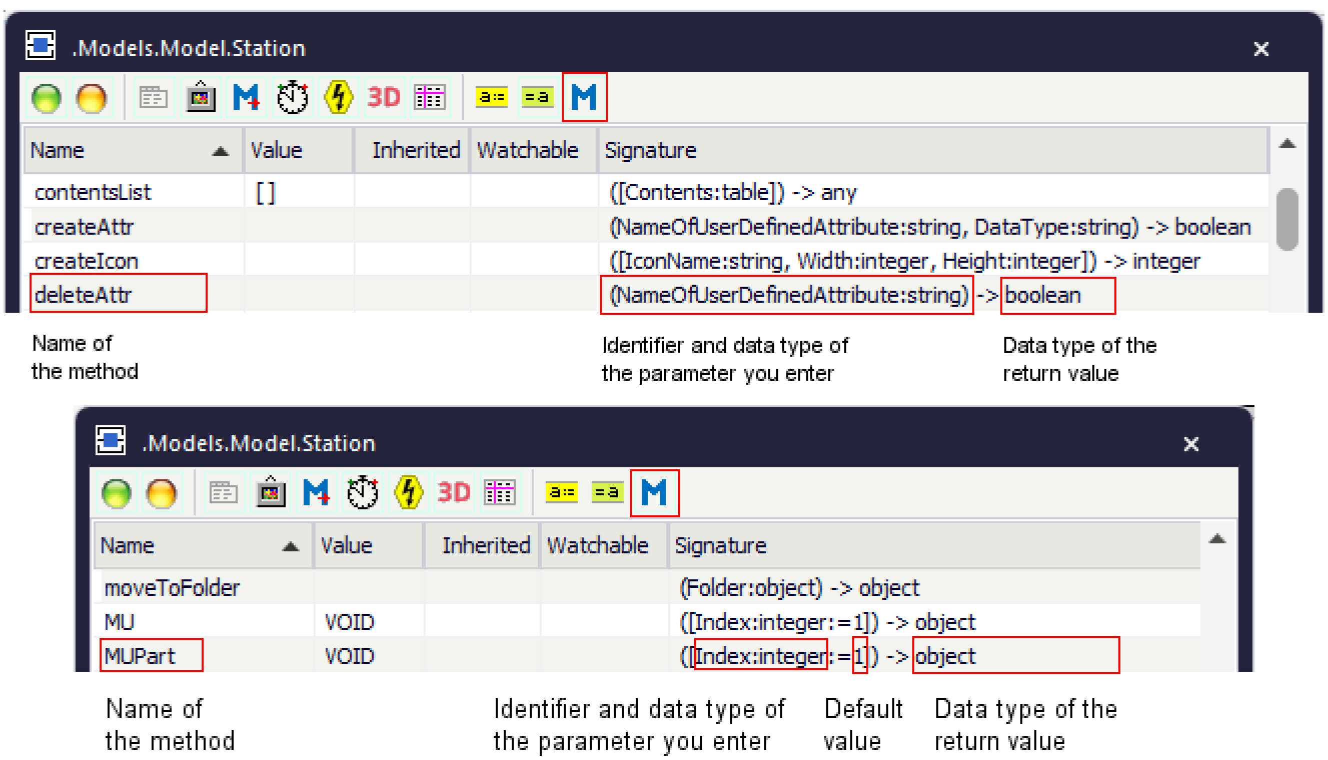
Methods of the ShiftCalendar

_Methods of the ShiftCalendar

The *ShiftCalendar* provides:

- The *methods* listed in the table of contents to the left.
- The **Methods of All Objects**.

To view all of the *methods*, *read-only attributes*, and *attributes* of the object, open the window **Show Attributes and Methods**. The figure below illustrates the information using the example of the object *Station*.



- Select **Show Attributes and Methods** on the context menu of the **Class Library** to show the *methods*, *read-only attributes*, and *attributes* of the selected **Class [general description]**.
- Press the **F8** key or click **Show Attributes and Methods** on the **Home** ribbon tab of the *Frame* into which you inserted an instance to show the *methods*, *read-only attributes*, and *attributes* of the selected **Instance [general description]**.

An example of the **Syntax** line of the individual methods might look like this:

```
<Path>.openDialog([CallOpenControl:boolean:=false]) → boolean
```

- The expression *<Path>* designates the path of the object to which the *method* applies.
- The signature of the method, consisting of the identifier and the data type of the parameter, is listed in parentheses. The expression *(Parameter:string)*, for example, designates a parameter of data type *string*. Instead of a constant value, you can also use a variable of the required type or a *method* that returns the required data type.

> **Note**
>
> Make sure to enter the parentheses for expressions within parentheses (...). Not entering them may lead to unexpected results and open the *Debugger*.

Optional parameters are listed within brackets. The expression *[,Parameter:boolean]*, for example, means that you can, but do not have to enter the *boolean* parameter.

If a parameter has a default value, the signature shows the default value after the parameter, `:= false` in the example above.

If the *method* has a return value, the signature shows its data type after the arrow ->, `→ boolean` in the example above.

calculateWorkingDuration [SimTalk]

Calculates the **Working Duration** between two points in time for the *ShiftCalendar* designated by <Path>.

Remarks

While calculating the working duration, *calculateWorkingDuration* takes pauses and shifts into account.

Type

Read-only attribute

Syntax

```
<Path>.calculateWorkingDuration(StartTime:dateTime, EndTime:dateTime) →
time
```

Parameters

You can specify the following parameters:

- The parameter *StartTime* of data type *dateTime* designates the start time from which on the *ShiftCalendar* computes the working duration.
- The parameter *EndTime* of data type *dateTime* designates the end time until which the *ShiftCalendar* computes the working duration.

Return Value

The return value has the data type *time*.

Example

```
var startDateTime   : dateTime
var endDateTime     : dateTime
var workingDuration : time
startDateTime := str_to_dateTime("02.01.2006 0:00")
endDateTime   := str_to_dateTime("02.01.2006 10:15")
workingDuration := MyShiftCalendar.calculateWorkingDuration(startDateTime,
endDateTime)
print workingDuration
```

See also

Working [state, material flow objects]

schedule [SimTalk] - ShiftCalendar

Schedules the date and time to start or to finish the production process for the *ShiftCalendar* designated by <Path>.

Remarks

The method *schedule* can schedule 100 years into the future.

Type

Method

Syntax

```
<Path>.schedule(StartTime:dateTime, Duration:time, Direction:string) →
dateTime
```

Parameters

You can specify the following parameters:

- The parameter *StartTime* of data type *dateTime* designates the start or end date and time from which on the *ShiftCalendar* schedules the production date.
- The parameter *Duration* of data type *time* designates the duration of the production process.
- The parameter *Direction* of data type *string* designates direction, i.e., if the *ShiftCalendar* computes `forward` or `backward` in time.
    - For `forward` scheduling, the production order requires a defined production time for manufacturing the product. Beginning at the start date, which normally is the active date, the method *schedule* computes the date on which the product has to be finished. It takes production time, shift times, weekends and holidays into consideration.
    - For `backward` scheduling, where you know the date on which the product has to available, as well as the production time, the method *schedule* computes the start date on which production has to start. It takes production time, shift times, weekends and holidays into consideration.

Return Value

The return value has the data type *dateTime*.

Example

```
startDateAndTime := str_to_dateTime("4.1.2003
9:00")                             // Friday 9:00
Duration :=
```

```
str_to_time("10:00:00.0")                                    // 10
hours
endDateAndTime := MyShiftCalendar.schedule(startDateAndTime,Duration,
"forward") // Monday 12:00
```

See also

[Schedule Date and Time to Start or Finish the Production Process]

Read-Only Attributes of the ShiftCalendar

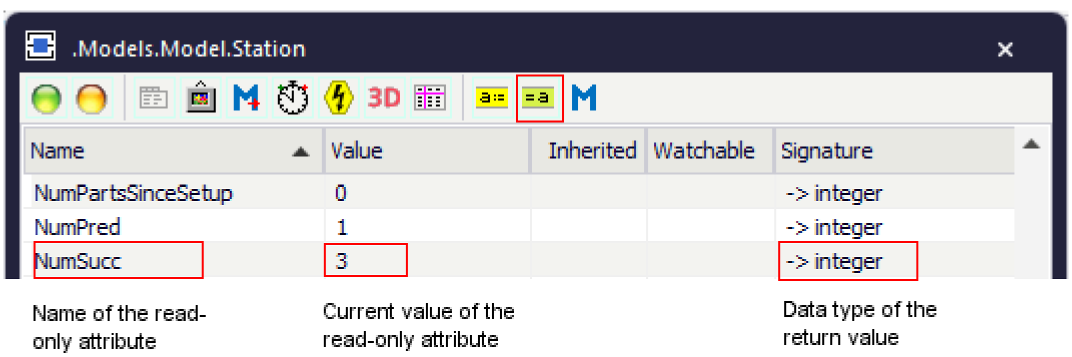
_Read-Only Attributes of the ShiftCalendar

The *ShiftCalendar* provides:

- The *read-only attributes* listed in the table of contents to the left.
- The **Read-Only Attributes of All Objects**.

You can query the values of the *read-only attributes*, but you cannot set them as *Plant Simulation* computes the value for the point-in-time at which you query it. In most cases a *read-only attribute* corresponds to an unavailable dialog item on one of the tabs of the object, for example on the tab **Statistics**.

To view all of the *methods*, *read-only attributes*, and *attributes* of the object, open the window **Show Attributes and Methods**. The figure below illustrates the information using the example of the object *Station*.



- Select **Show Attributes and Methods** on the context menu of the **Class Library** to show the *methods*, *read-only attributes*, and *attributes* of the selected **Class [general description]**.
- Press the **F8** key or click **Show Attributes and Methods** on the **Home** ribbon tab of the *Frame* into which you inserted an instance to show the *methods*, *read-only attributes*, and *attributes* of the selected **Instance [general description]**.